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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (CSE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CSE302 - DEEP LEARNING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define a biological neural network and an artificial neural network. How do they differ?	10	L*2	1	1,2
	b) Explain the perceptron architecture and its role in neural networks.	10	L2	1	1,2
2.	a) Describe the basic structure of a feed-forward neural network.	10	L2	2	1,2
	b) Design a NAND gate with Mc Culloch - Pitts neuron.	10	L3	2	1,2,3
3.	a) Explain the various applications of Artificial Neural Networks (ANNs) in real-world problem. Give specific examples in areas such as image recognition, natural language processing and medical diagnosis.	10	L1	2	1
	b) Define and explain the purpose of activation functions in Neural networks. Compare and contrast different activation functions, such as Sigmoid, Tanh, and ReLU, highlighting their strengths and weaknesses.	10	L1	3	1
Unit – II					
4.	a) Show how Back propagation learns: learning via gradient descent.	10	L3	3	1,2,3
	b) List the key differences between optimization algorithms SGD, AdaGrad, RMSProp, and Adam.	10	L3	3	1,2,3
5.	a) Show how to solve XOR problem using feedforward neural networks.	10	L3	4	1,2,3
	b) With an example show working of CNN operations. i) Padding ii) Pooling	10	L3	3	1,2,3
6.	a) Discuss the properties of CNN representations.	10	L3	4	1,2,3
	b) Describe any five real world applications of convolutional neural networks.	10	L3	5	1,2,3
Unit – III					
7.	a) With a diagram explain Deep Recurrent Networks.	10	L2	4	1,2
	b) Discuss & Compare CNN and RNN.	10	L3	5	1,2,3
8.	a) Illustrate the design of a CNN for image classification using Keras. Provide the key steps in the implementation.	10	L3	5	1,2
	b) Differentiate between object localization and object detection. How does CNN architecture vary for each task?	10	L3	5	1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
